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$$\sum_{n=0}^{\infty} e^{-\frac{n^2}{2}}$$

Criterium 7(+) (d'Alembert)

$$\text{Stel } x_n = e^{-\frac{n^2}{2}}$$

$$\text{Stel } x_{n+1} = e^{-\frac{(n+1)^2}{2}}$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{e^{-\frac{(n+1)^2}{2}}}{e^{-\frac{n^2}{2}}} \\ &= \lim_{n \rightarrow \infty} e^{-\frac{n^2+2n+1}{2} + \frac{n^2}{2}} \\ &= \lim_{n \rightarrow \infty} e^{\frac{-n^2-2n-1+n^2}{2}} \\ &= \lim_{n \rightarrow \infty} e^{\frac{-2n-1}{2}} \\ &= \lim_{n \rightarrow \infty} e^{-n-\frac{1}{2}} = 0 \end{aligned}$$

Aangezien $L = 0 < 1$, is de reeks volgens het criterium van d'Alembert convergent.

References